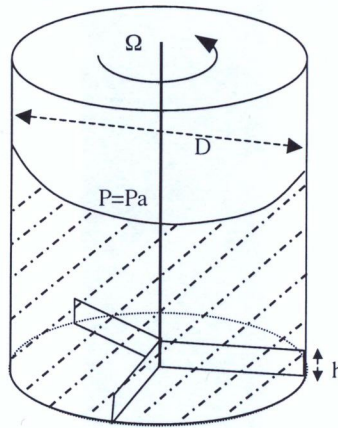


A large container of diameter $D = 10$ m contains a volume $V = 1000$ m³ of tar ($\nu_t = 10^{-3}$ m²/s, $\rho_t = 2 \times 10^3$ kg/m³). To avoid solidification, the tar is continuously moved with a 3-blade stirrer of height $h = 1$ m placed at its bottom, which rotates with angular velocity $\Omega = 1$ s⁻¹, as shown in the figure. Design a small experiment with a geometrically similar model to determine the power W needed to operate the stirrer. In particular,

- Use the Π theorem to reduce the parametric dependence of W .
- By using water in the experiment as working fluid, obtain the values of D_e , h_e , V_e y Ω_e required to guarantee that the experiment is physically similar to the prototype.
- Obtain the ratio W/W_e between the power measured in the experiment W_e and that required to move the stirrer of the real prototype.



$$W = W(\Omega, g, h, D, \nu, \rho, \mu) \Rightarrow \frac{W}{\mu \Omega^2 h^3} = f\left(\frac{g}{\Omega^2 h}, \frac{D}{h}, \frac{V}{h^3}, \frac{\rho \Omega h^2}{\mu}\right)$$

$$\left. \begin{aligned} \frac{g}{\Omega^2 h} &= \frac{g}{\Omega_e^2 h_e} \Rightarrow \frac{h_e}{h} = \left(\frac{\Omega}{\Omega_e}\right)^2 \\ \frac{\Omega h^2}{\nu_t} &= \frac{\Omega_e h_e^2}{\nu_w} \rightarrow \left(\frac{h_e}{h}\right)^2 = \left(\frac{\Omega}{\Omega_e}\right) \frac{\nu_w}{\nu_t} \end{aligned} \right\} \quad \begin{aligned} \frac{h_e}{h} &= \left(\frac{\nu_t}{\nu_w}\right)^{-2/3} = 0.01 \rightarrow \boxed{h_e = 1 \text{ cm}} \\ \frac{\Omega_e}{\Omega} &= \left(\frac{\nu_t}{\nu_w}\right)^{1/3} = 10 \rightarrow \boxed{\Omega_e = 10 \text{ s}^{-1}} \end{aligned}$$

$$\frac{D}{h} = \frac{D_e}{h_e} \rightarrow \boxed{D_e = \frac{h_e}{h} D = 10 \text{ cm}} \quad \frac{V}{h^3} = \frac{V_e}{h_e^3} \rightarrow \boxed{V_e = \left(\frac{h_e}{h}\right)^3 V = 1 \text{ litre}}$$

$$\frac{W}{W_e} = \frac{\mu_t \Omega^2 h^3}{\mu_w \Omega_e^2 h_e^3} = \left(\frac{\rho_t}{\rho_w}\right) \left(\frac{\nu_t}{\nu_w}\right) \left(\frac{\Omega}{\Omega_e}\right)^2 \left(\frac{h}{h_e}\right)^3 = 2 \times 10^7$$